

Dataframe Basics

We've learned about vectors and their two-dimensional counterpart, matrices. Now we will learn about Dataframes, one of the main tools for data analysis with R! Matrix inputs were limited because all the data inside of the matrix had to be of the same data type (numerics, logicals, etc). With Dataframes we will be able to organize and mix data types to create a very powerful data structure tool!

R actually has built in DataFrames for quick reference to play around with! Check out the following dataframes that are built-in!

In [2]:

```
# Dataframe about states
state.x77
```

Out[2]:

	Population	Income	Illiteracy	Life Exp	Murder	HS Grad	Frost	Area
Alabama	3615.00	3624.00	2.10	69.05	15.10	41.30	20.00	50708.00
Alaska	365.00	6315.00	1.50	69.31	11.30	66.70	152.00	566432.00
Arizona	2212.00	4530.00	1.80	70.55	7.80	58.10	15.00	113417.00
Arkansas	2110.00	3378.00	1.90	70.66	10.10	39.90	65.00	51945.00
California	21198.00	5114.00	1.10	71.71	10.30	62.60	20.00	156361.00
Colorado	2541.00	4884.00	0.70	72.06	6.80	63.90	166.00	103766.00
Connecticut	3100.00	5348.00	1.10	72.48	3.10	56.00	139.00	4862.00
Delaware	579.00	4809.00	0.90	70.06	6.20	54.60	103.00	1982.00
Florida	8277.00	4815.00	1.30	70.66	10.70	52.60	11.00	54090.00
Georgia	4931.00	4091.00	2.00	68.54	13.90	40.60	60.00	58073.00
Hawaii	868.0	4963.0	1.9	73.6	6.2	61.9	0.0	6425.0
Idaho	813.00	4119.00	0.60	71.87	5.30	59.50	126.00	82677.00
Illinois	11197.00	5107.00	0.90	70.14	10.30	52.60	127.00	55748.00
Indiana	5313.00	4458.00	0.70	70.88	7.10	52.90	122.00	36097.00
Iowa	2861.00	4628.00	0.50	72.56	2.30	59.00	140.00	55941.00
Kansas	2280.00	4669.00	0.60	72.58	4.50	59.90	114.00	81787.00
Kentucky	3387.0	3712.0	1.6	70.1	10.6	38.5	95.0	39650.0
Louisiana	3806.00	3545.00	2.80	68.76	13.20	42.20	12.00	44930.00
Maine	1058.00	3694.00	0.70	70.39	2.70	54.70	161.00	30920.00
Maryland	4122.00	5299.00	0.90	70.22	8.50	52.30	101.00	9891.00
Massachusetts	5814.00	4755.00	1.10	71.83	3.30	58.50	103.00	7826.00
Michigan	9111.00	4751.00	0.90	70.63	11.10	52.80	125.00	56817.00
Minnesota	3921.00	4675.00	0.60	72.96	2.30	57.60	160.00	79289.00
Mississippi	2341.00	3098.00	2.40	68.09	12.50	41.00	50.00	47296.00
Missouri	4767.00	4254.00	0.80	70.69	9.30	48.80	108.00	68995.00
Montana	746.00	4347.00	0.60	70.56	5.00	59.20	155.00	145587.00
Nebraska	1544.0	4508.0	0.6	72.6	2.9	59.3	139.0	76483.0
Nevada	590.00	5149.00	0.50	69.03	11.50	65.20	188.00	109889.00
New Hampshire	812.00	4281.00	0.70	71.23	3.30	57.60	174.00	9027.00
New Jersey	7333.00	5237.00	1.10	70.93	5.20	52.50	115.00	7521.00
New Mexico	1144.00	3601.00	2.20	70.32	9.70	55.20	120.00	121412.00

New York	18076.00	4903.00	1.40	70.55	10.90	52.70	82.00	47831.00
Population								Area
North Carolina	5441.00	3875.00	1.80	69.21	11.10	38.50	80.00	48798.00
North Dakota	637.00	5087.00	0.80	72.78	1.40	50.30	186.00	69273.00
Ohio	10735.00	4561.00	0.80	70.82	7.40	53.20	124.00	40975.00
Oklahoma	2715.00	3983.00	1.10	71.42	6.40	51.60	82.00	68782.00
Oregon	2284.00	4660.00	0.60	72.13	4.20	60.00	44.00	96184.00
Pennsylvania	11860.00	4449.00	1.00	70.43	6.10	50.20	126.00	44966.00
Rhode Island	931.0	4558.0	1.3	71.9	2.4	46.4	127.0	1049.0
South Carolina	2816.00	3635.00	2.30	67.96	11.60	37.80	65.00	30225.00
South Dakota	681.00	4167.00	0.50	72.08	1.70	53.30	172.00	75955.00
Tennessee	4173.00	3821.00	1.70	70.11	11.00	41.80	70.00	41328.00
Texas	12237.0	4188.0	2.2	70.9	12.2	47.4	35.0	262134.0
Utah	1203.0	4022.0	0.6	72.9	4.5	67.3	137.0	82096.0
Vermont	472.00	3907.00	0.60	71.64	5.50	57.10	168.00	9267.00
Virginia	4981.00	4701.00	1.40	70.08	9.50	47.80	85.00	39780.00
Washington	3559.00	4864.00	0.60	71.72	4.30	63.50	32.00	66570.00
West Virginia	1799.00	3617.00	1.40	69.48	6.70	41.60	100.00	24070.00
Wisconsin	4589.00	4468.00	0.70	72.48	3.00	54.50	149.00	54464.00
Wyoming	376.00	4566.00	0.60	70.29	6.90	62.90	173.00	97203.00

In [8]:

```
# US personal expense
USPersonalExpenditure
```

Out[8]:

	1940	1945	1950	1955	1960
Food and Tobacco	22.2	44.5	59.6	73.2	86.8
Household Operation	10.5	15.5	29.0	36.5	46.2
Medical and Health	3.53	5.76	9.71	14.00	21.10
Personal Care	1.04	1.98	2.45	3.40	5.40
Private Education	0.341	0.974	1.800	2.600	3.640

In [4]:

```
# Women
women
```

Out[4]:

	height	weight
1	58	115
2	59	117
3	60	120
4	61	123
5	62	126
6	63	129
7	64	132

8	65	135
9	66	139
10	67	142
11	68	146
12	69	150
13	70	154
14	71	159
15	72	164

To get a list of all available built-in dataframes, use `data()`

In [7]:

```
data()
```

Working with DataFrames

You'll notice the states dataframe was really big, we can use the `head()` and `tail()` functions to view the first and last 6 rows respectively. Let's take a look:

In [9]:

```
# Quick variable assignment to save typing
states <- state.x77
```

In [11]:

```
head(states)
```

Out[11]:

	Population	Income	Illiteracy	Life Exp	Murder	HS Grad	Frost	Area
Alabama	3615.00	3624.00	2.10	69.05	15.10	41.30	20.00	50708.00
Alaska	365.00	6315.00	1.50	69.31	11.30	66.70	152.00	566432.00
Arizona	2212.00	4530.00	1.80	70.55	7.80	58.10	15.00	113417.00
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California	21198.00	5114.00	1.10	71.71	10.30	62.60	20.00	156361.00
Colorado	2541.00	4884.00	0.70	72.06	6.80	63.90	166.00	103766.00

In [12]:

```
tail(states)
```

Out[12]:

	Population	Income	Illiteracy	Life Exp	Murder	HS Grad	Frost	Area
Vermont	472.00	3907.00	0.60	71.64	5.50	57.10	168.00	9267.00
Virginia	4981.00	4701.00	1.40	70.08	9.50	47.80	85.00	39780.00
Washington	3559.00	4864.00	0.60	71.72	4.30	63.50	32.00	66570.00
West Virginia	1799.00	3617.00	1.40	69.48	6.70	41.60	100.00	24070.00
Wisconsin	4589.00	4468.00	0.70	72.48	3.00	54.50	149.00	54464.00
Wyoming	376.00	4566.00	0.60	70.29	6.90	62.90	173.00	97203.00

	Population	Income	Illiteracy	Life Exp	Murder	HS Grad	Frost	Area
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DataFrames - Overview of information

We can use the **str()** to get the structure of a dataframe, which gives information on the structure of the dataframe and the data it contains, such as variable names and data types. We can use **summary()** to get a quick statistical summary of all the columns of a DataFrame, depending on the data, this may or may not be useful!

In [13]:

```
# Statistical summary of data
summary(states)
```

Out[13]:

Population	Income	Illiteracy	Life Exp
Min. : 365	Min. :3098	Min. :0.500	Min. :67.96
1st Qu.: 1080	1st Qu.:3993	1st Qu.:0.625	1st Qu.:70.12
Median : 2838	Median :4519	Median :0.950	Median :70.67
Mean : 4246	Mean :4436	Mean :1.170	Mean :70.88
3rd Qu.: 4968	3rd Qu.:4814	3rd Qu.:1.575	3rd Qu.:71.89
Max. :21198	Max. :6315	Max. :2.800	Max. :73.60

Murder	HS Grad	Frost	Area
Min. : 1.400	Min. :37.80	Min. : 0.00	Min. : 1049
1st Qu.: 4.350	1st Qu.:48.05	1st Qu.: 66.25	1st Qu.: 36985
Median : 6.850	Median :53.25	Median :114.50	Median : 54277
Mean : 7.378	Mean :53.11	Mean :104.46	Mean : 70736
3rd Qu.:10.675	3rd Qu.:59.15	3rd Qu.:139.75	3rd Qu.: 81162
Max. :15.100	Max. :67.30	Max. :188.00	Max. :566432

In [14]:

```
# Structure of Data
str(states)
```

```
num [1:50, 1:8] 3615 365 2212 2110 21198 ...
- attr(*, "dimnames")=List of 2
 ..$ : chr [1:50] "Alabama" "Alaska" "Arizona" "Arkansas" ...
 ..$ : chr [1:8] "Population" "Income" "Illiteracy" "Life Exp" ...
```

Creating Data frames

A quick note some people write Dataframe as one word, but in R its more commonly written as two words: data frame. Not a very huge deal either way, but if someone writes DataFrame they may be referring to a Python/pandas DataFrame, so keep that in mind!

We can create data frames using the **data.frame()** function and pass vectors as arguments, which will then convert the vectors into columns of the data frame. Let's see a simple example:

In [16]:

```
# Some made up weather data
days <- c('mon', 'tue', 'wed', 'thu', 'fri')
temp <- c(22.2, 21, 23, 24.3, 25)
rain <- c(TRUE, TRUE, FALSE, FALSE, TRUE)
```

In [18]:

```
# Pass in the vectors:
df <- data.frame(days, temp, rain)
```

In [19]:

```
df
```

Out

Out[19]:

	days	temp	rain
1	mon	22.2	TRUE
2	tue	21	TRUE
3	wed	23	FALSE
4	thu	24.3	FALSE
5	fri	25	TRUE

In [20]:

```
# Check structure  
str(df)
```

```
'data.frame': 5 obs. of 3 variables:  
 $ days: Factor w/ 5 levels "fri","mon","thu",...: 2 4 5 3 1  
 $ temp: num 22.2 21 23 24.3 25  
 $ rain: logi TRUE TRUE FALSE FALSE TRUE
```

In [21]:

```
summary(df)
```

Out[21]:

```
   days      temp      rain  
fri:1  Min.    :21.0  Mode :logical  
mon:1  1st Qu.:22.2  FALSE:2  
thu:1  Median :23.0   TRUE :3  
tue:1  Mean   :23.1  NA's :0  
wed:1  3rd Qu.:24.3  
       Max.    :25.0
```

That's it for the basics, up next we will learn about selection and indexing Data Frame elements!